

The empirical recurrence for OEIS A235852: recovering a conjecture that is not written down

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7 September 2026

Abstract

OEIS A235852 records “Empirical recurrence of order 43”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 72$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 43, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 43 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A235852 is “Number of $(n+1) \times (3+1)$ 0..3 arrays with the upper median equal to the lower median in every 2×2 subblock.”. It has offset 1 and begins

4524, 121244, 2848572, 73075844, 1811192156, 45872598284, 1149967012292, 29010506163148, ...

The entry states:

Empirical recurrence of order 43 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:53:43, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 72$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 144$ exact terms of the model returns order 43 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{43} c_i a(n-i),$$

c_1	10	c_{12}	−31910038666038	c_{23}	−153102745476693032220	c_{34}	155752880704010
c_2	938	c_{13}	87022824438571	c_{24}	−908382500169585923210	c_{35}	−14714832080351
c_3	−6069	c_{14}	1222535630885288	c_{25}	1093841149339073577652	c_{36}	−12340194607009
c_4	−336464	c_{15}	−2910179495235194	c_{26}	5257160127720059582280	c_{37}	115940116231162
c_5	1659902	c_{16}	−33956393252322906	c_{27}	−5785493956439619168320	c_{38}	55021720302466
c_6	64617998	c_{17}	70076545890198185	c_{28}	−21657253726079257045984	c_{39}	−5230286702935
c_7	−264646990	c_{18}	690723517863448433	c_{29}	22339913267145814161408	c_{40}	−1157003150272
c_8	−7615657846	c_{19}	−1231099569715279265	c_{30}	62262411558655923269376	c_{41}	11540580293454
c_9	26965762802	c_{20}	−10328781616587421884	c_{31}	−61496926819617770929408	c_{42}	8096125956533
c_{10}	592898503308	c_{21}	15938066264431869169	c_{32}	−121674023542233990980096	c_{43}	−914248732295
c_{11}	−1841818545841	c_{22}	113385092806236901886	c_{33}	116874659806551059243008		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 43, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $43 + 4$ terms removed returns order 43 as well, so $\deg q_{\min} = 43 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $43 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 43 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A235852.

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